

Vermal Cerebellum Differences between 3T and 7T fMRI within Subclinical Anxiety and Healthy Adults

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Introduction

The cerebellum, traditionally viewed as a motor region, has received sparse functional MRI (fMRI) research attention in comparison to classical cerebral regions. However, in the past two decades, studies have shown its involvement in cognition and affect^{1,2}. Additionally, increased recognition has been given to the cerebellar vermis and its potential role in psychopathology like anxiety³.

Aims & Hypotheses

The present study aims to examine the connectivity differences between 7T and 3T fMRI data, as well as the role the vermis has in anxiety.

Hypothesis 1: Sub-clinical anxiety participants will have altered vermal connectivity compared to healthy controls.

Hypothesis 2: 7T vermal connectivity will be stronger than 3T connectivity.

Hypothesis 3: The dominant pattern of covariation in posterior vermis connectivity will predict anxiety severity.

Methods

Sample: Participants were from the Welsh Advanced Neuroimaging Database⁴. The sample included those who completed 3T, 7T, and the Generalized Anxiety Disorder-7 (GAD-7) survey⁵. Subclinical anxiety presence was determined by whether one’s total GAD-7 score was ≥ 5 .

Group	Total Sample	Sub-clinical Anxiety (Anx)	Healthy Control (HC)
Sample (n)	109	29	80
Sex (M/F)	38/71	10/19	28/52
Age Range (Mean)	18-63 (29.4)	18-57 (26.3)	18-63 (30.6)

rs-fMRI Acquisition: All participants were instructed to focus on a white fixation cross overlaid on a black background for a continuous scan. 3T Data were acquired on 3T PRISMA Scanner and 7T Data were acquired on 7T Siemens Magnetom Scanner.

Preprocessing: Functional and anatomical data were preprocessed via CONN Toolbox 22.v2407⁶ using the *default_mni* pipeline. Functional data was left unsmoothed. Follow-up analyses for 7T data used were conducted using HALFpipe (v1.3.2)⁷ with Singularity (v1.3.6). Data were smoothed to 6 mm FWHM, and atlas-based functional connectivity was extracted using the Brainnetome and Buckner cerebellar atlases^{8,9}. Quality control followed ENIGMA-HALFpipe guidelines.

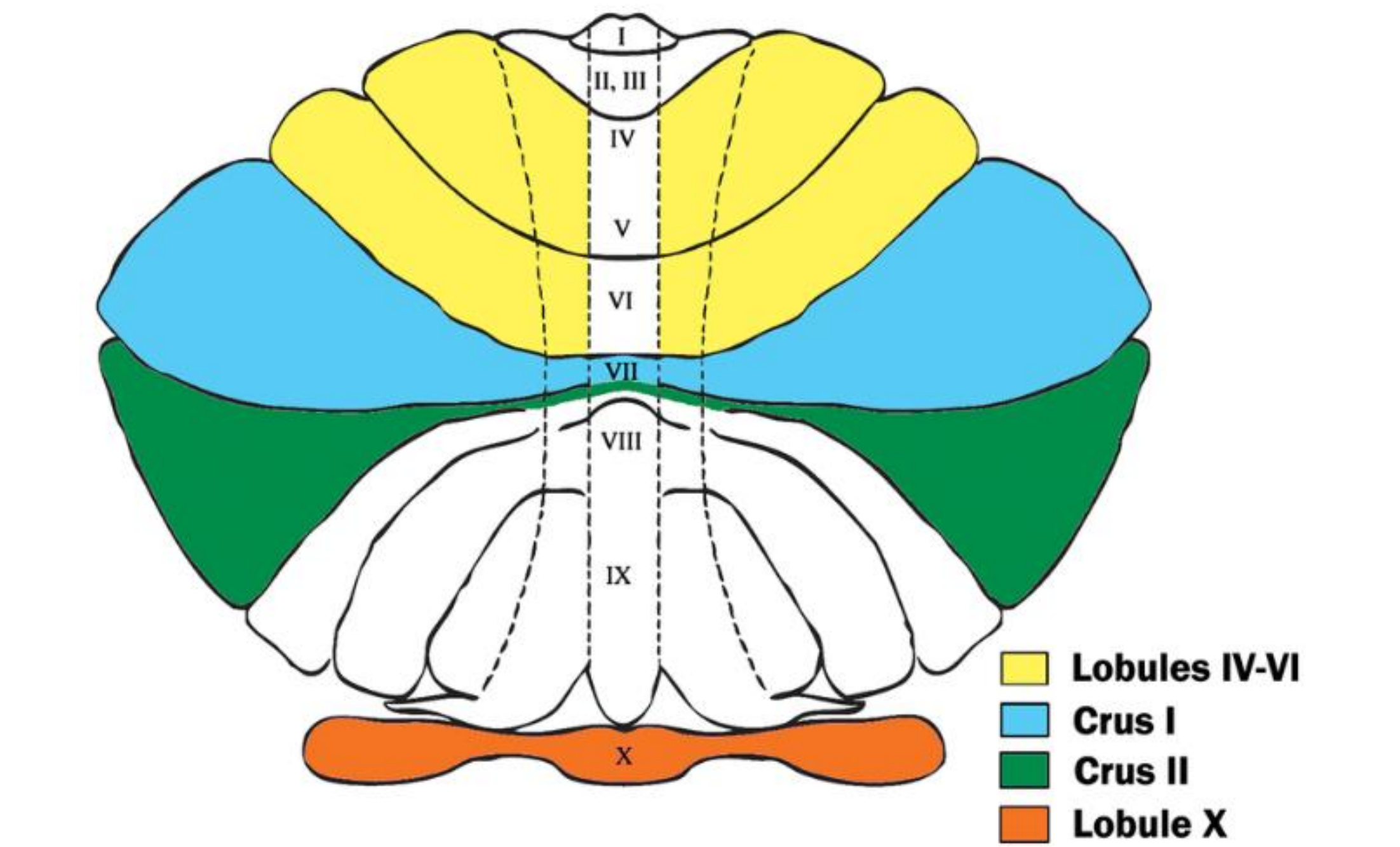


Figure 1. Parcellated Cerebellum flat map. Vermal regions (I-X) are depicted as midline structures labeled via roman numerals¹⁰.

Results

Between Groups Analyses:

Posterior Vermal ROIs: Vermis 6, Vermis 7, Vermis 8, and Vermis 9

1st Level Analyses: Seed-based connectivity maps were calculated for posterior vermal seeds from the combined default Harvard-Oxford and AAL Atlases in CONN.

2nd Level Analyses: Using a General Linear Model, while covarying for age and sex, differences between anxious and healthy groups were performed for each seed. **No significant differences** were seen for Anx>HC or HC>Anxiety.

3T versus 7T Analyses:

2nd Level Analyses: Connectivity maps for each vermal seed were taken from CONN and analyzed in SPM12 via a paired *t*-test for within group effects to examine differences in 3T and 7T connectivity.

7T>3T Results: Significant clusters (pFWE<0.05, cluster extent ≥ 20 voxels) between HC only scans, across all vermal seeds were observed. When testing for connectivity differences only in the anxiety subsample, effects were confined to the predefined seed region for Vermis 9.

Table 1. Regions showing significantly greater activation for 7T>3T in HC.

Seed Region	k _E	MNI (x,y,z)	p (FWE corr.)
Vermis 6	1325	4, -72, -12	P $\leq$ 0.001
	520	0, -72, 4	
	71	2, -66, -34	
	51	2, 0, 54	
	38	-32, 6, 8	
	26	2, 12, 36	
	24	4, -2, 66	
	22	8, -62, -50	
	21	36, -14, 40	
Vermis 7	1654	4, -74, -26	P $\leq$ 0.001
Vermis 8	413	0, -60, -36	P $\leq$ 0.001
	70	0, -48, -16	
	25	0, 14, 38	
	21	26, -42, -30	
Vermis 9	234	2, -54, -34	P $\leq$ 0.001
	38	-10, -16, 6	
	24	-14, 2, 14	
	20	12, -14, 4	

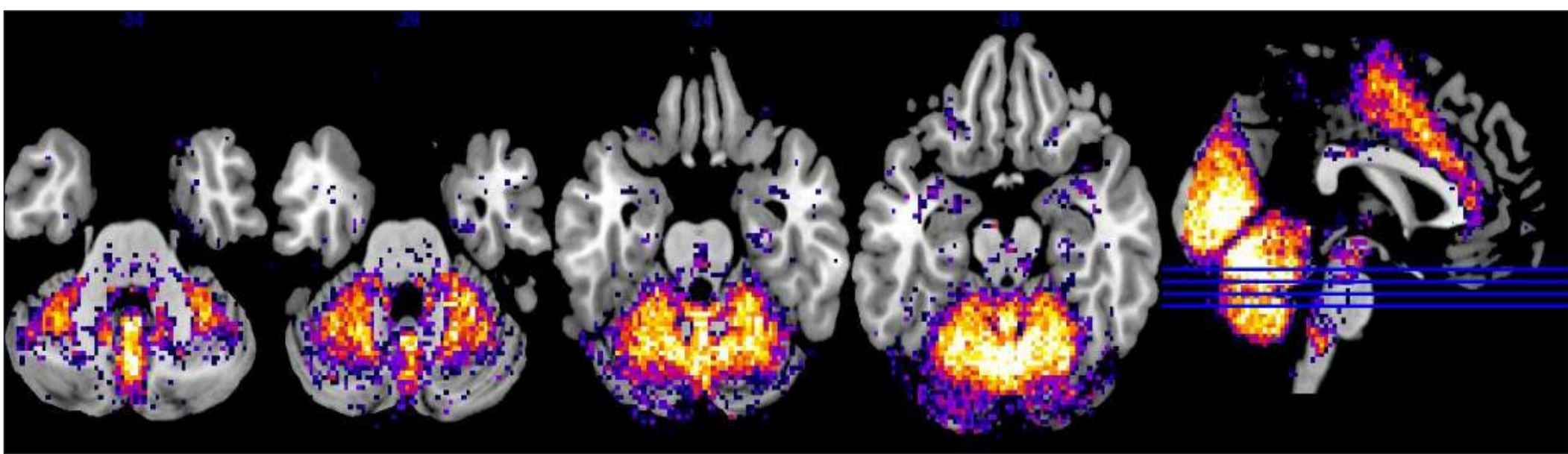


Figure 2. Multi-slice axial view of Vermis 6 seed 7T connectivity in HC, threshold $z > 3$.

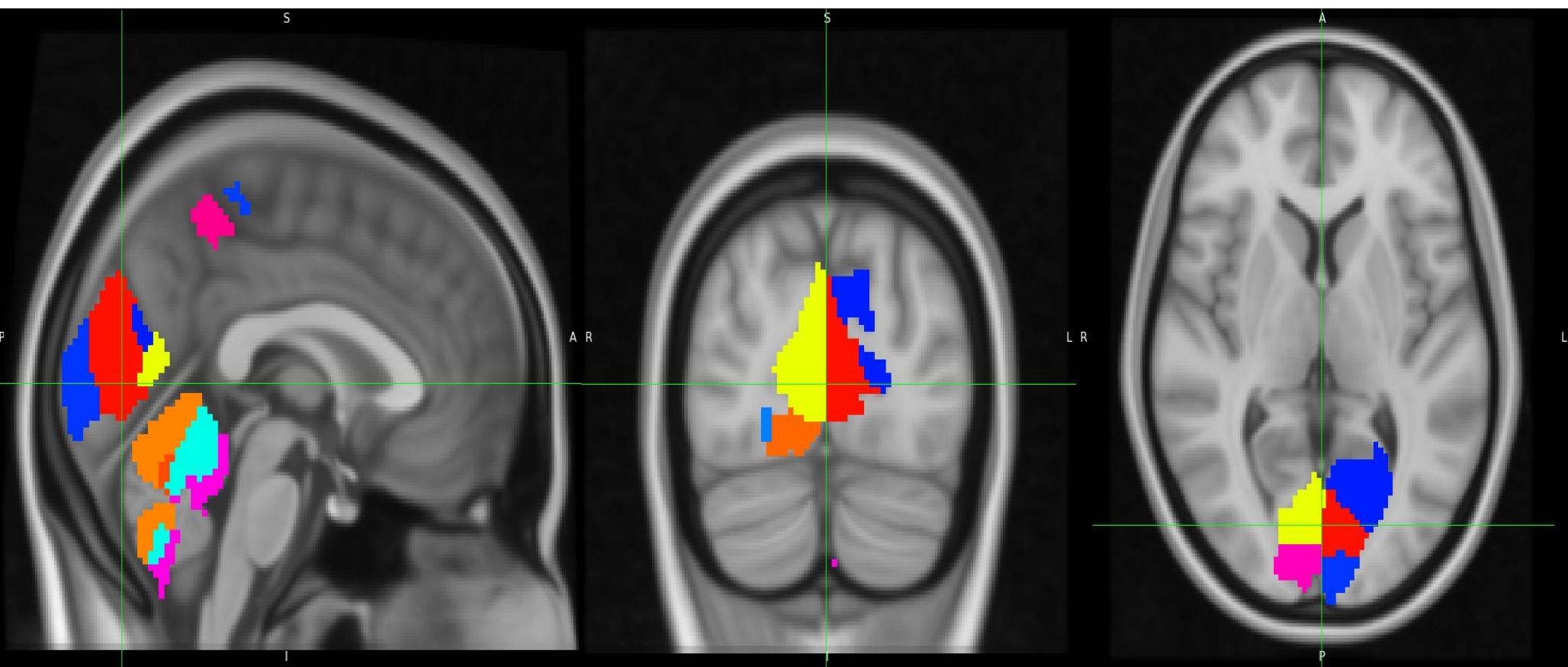


Figure 3. Top 15 functional connections for Buckner Network 2. ROIs are defined by the Brainnetome and Buckner Network atlases. Regions were selected based on the highest mean Fisher *z*-transformed connectivity (threshold: $z > 1.4$). Regions are listed in sequential order of mean connectivity magnitude: Buckner Network 7, Buckner Network 9, Buckner Network 4, Buckner Network 3, Buckner Network 15, Right Rostral Lingual Gyrus, Right Caudal Cuneus Gyrus, Right Rostral Cuneus Gyrus, Left Caudal Cuneus Gyrus, Right Parietal Medial Area 5, Left Rostral Lingual Gyrus, Left Rostral Cuneus Gyrus, Right Paracentral Lobule Area 1–3, Right Caudal Lingual Gyrus, and Left Ventromedial Parieto-occipital Sulcus.

Linear regression for seed-to-atlas functional connectivity matrix:

Seed: Network 2 (Vermis 6) of Buckner 17-network cerebellum parcellation

Regression Equation: Anxiety $\sim$ PC1 + Age + Sex + FD

Results: No Significance

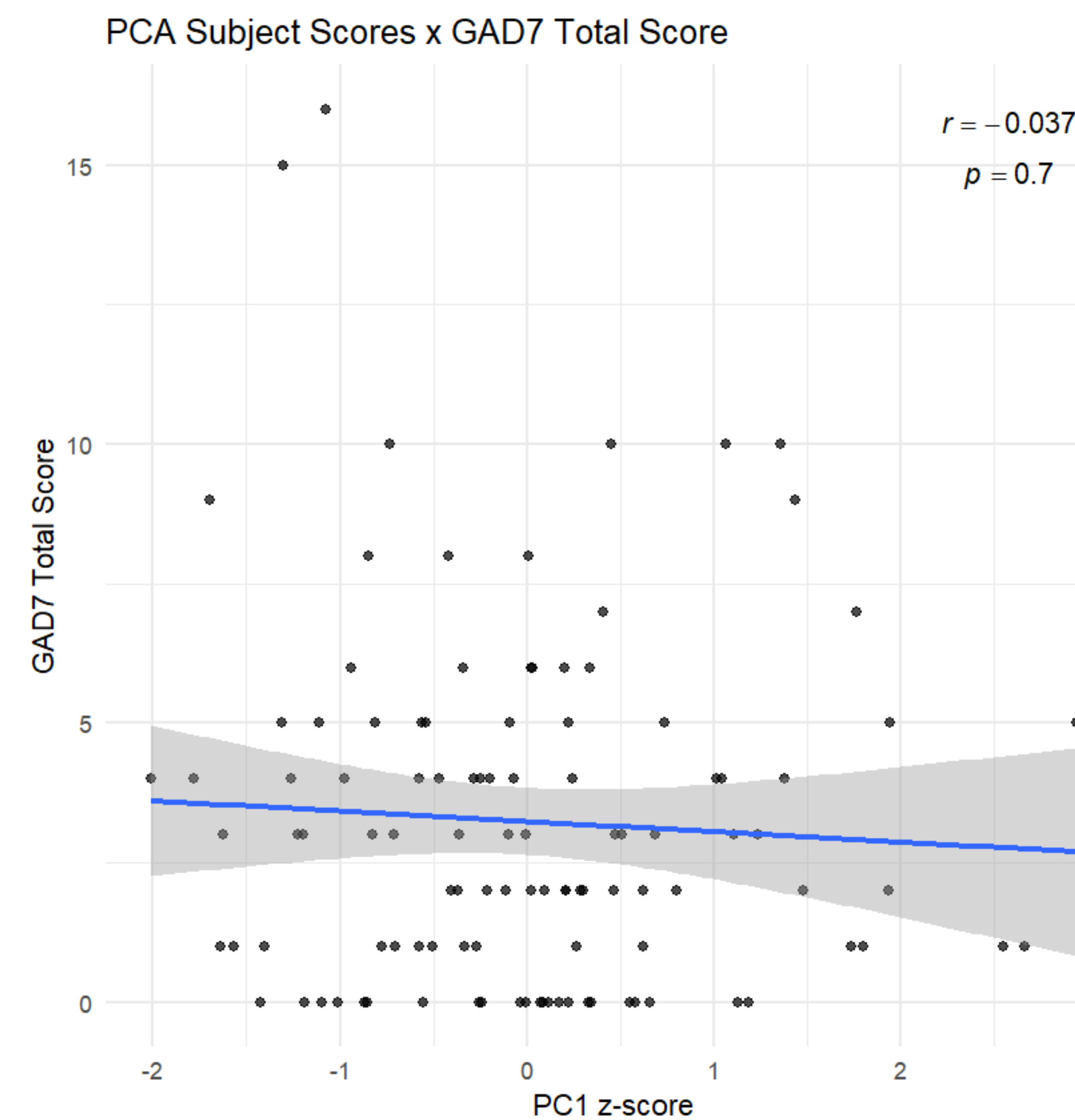


Figure 4. Scatterplot showing the linear relationship between Principal Component 1 (PC1) and anxiety scores, with fitted regression line.

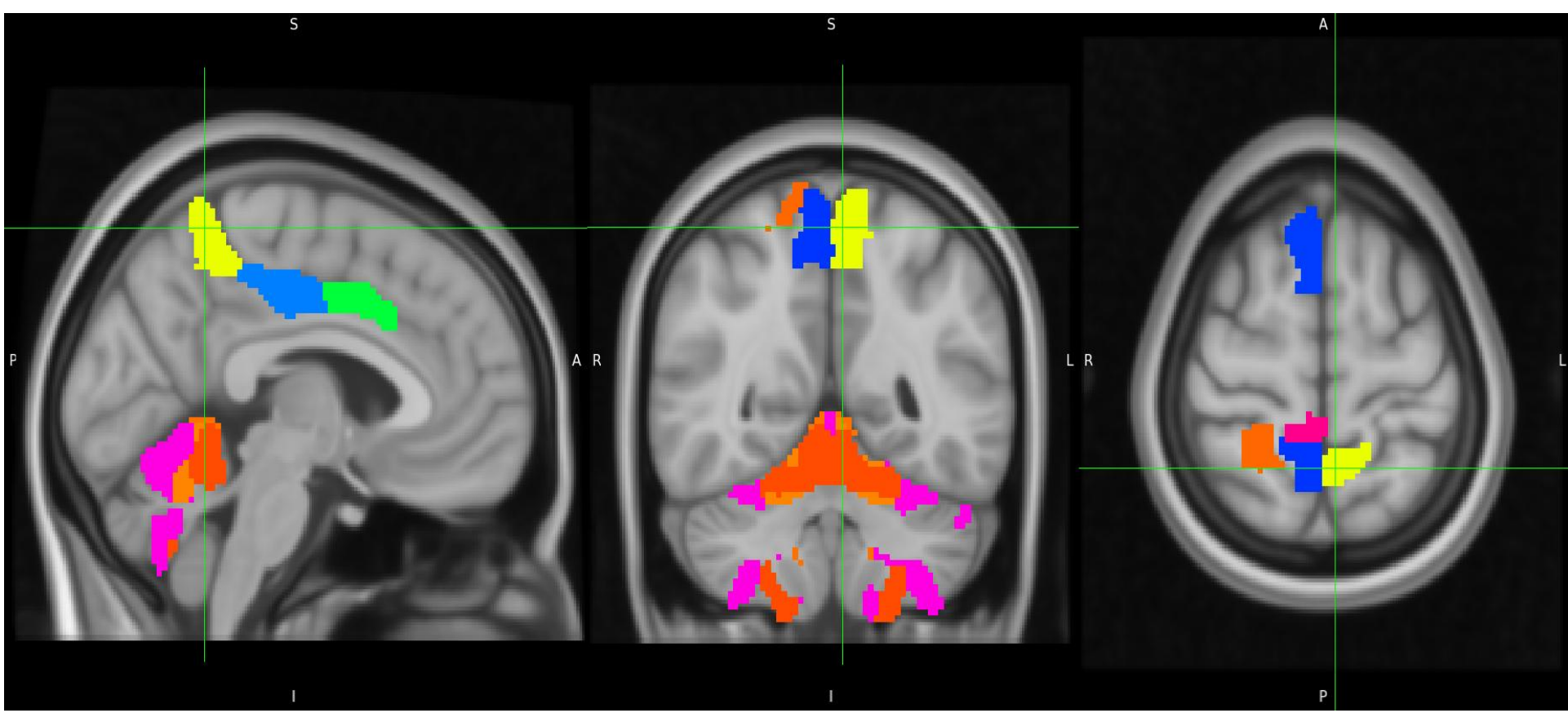


Figure 5. Dominant seed-to-atlas pairs contributing to PC1 Buckner Network 2 connectivity. The regions are: Buckner Network 3, Buckner Network 4, Buckner Network 7, Buckner Network 15, Right Precuneus Medial Area 5, Left Precuneus Medial Area 5, Right Postcentral Area 7, Left Caudal Area 23, Right Caudal Area 23, Right Paracentral Lobule Area 1-3, Right Posterior Parietal Thalamus, Right Superior Frontal Gyrus- Medial Area 8, Left Inferior Parietal Lobe- Rostroventral Area 40, Left Postcentral Gyrus Area 1-3, and Left Cingulate Gyrus- Caudo-dorsal Area 24.

Conclusions & Future Directions

Between group comparisons between Subclinical Anxiety and Healthy Controls were insignificant.

7T>3T showed significant differences in cerebral connectivity in the thalamus, caudate, cingulum, insula, supplementary motor area, and the precentral gyrus in healthy controls.

Increased sensitivity to cerebello-cortical and subcortical connectivity is seen in 7T data.

Vermal connectivity did not predict anxiety severity.

Vermal areas are connected to emotional processing and salience network regions.

Future work should use enhanced parcellated cerebellar maps, as well as clinical anxiety samples.

Additionally, analyses utilizing effective connective, rather than static connectivity could further highlight the role of the cerebellum has cognitive and affective disorders.

References



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